Min Max Stack Construction

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The problem

- A stack with O(1) push, pop, peek, getMin and getMax by storing the running min/max alongside each element.

Intuition

- Store the current min and max alongside each pushed value so they're always available.

Approach

- 1. Push value with updated min and max
- 2. Pop removes both
- 3. Peek, getMin and getMax read the top

Java solution

```java
public class MinMaxStackConstruction {
    public static class MinMaxStack {
        private final List<int[]> st = new ArrayList<>();

        public int peek() { return st.get(st.size() - 1)[0]; }
        public int pop() { return st.remove(st.size() - 1)[0]; }
        public void push(int v) {
            if (st.isEmpty()) st.add(new int[]{v, v, v});
            else { int[] t = st.get(st.size() - 1); st.add(new int[]{v, Math.min(v, t[1]), Math.max(v, t[2])}); }
        }
        public int getMin() { return st.get(st.size() - 1)[1]; }
        public int getMax() { return st.get(st.size() - 1)[2]; }
    }
}
```

Complexity

- Time: O(1) per op
- Space: O(n)

Recap

- Store the current min and max alongside each pushed value so they're always available.
- Complexity: O(1) per op